Embedded systems Architecture (BACSE324) Assessment 8: Hardware Interfacing of Led, Servo Motor, Ultrasonic Sensor and LCD with Arduino Uno

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<tr><td>Lab Sl. No.</td><td>8</td><td rowspan="5">  </td></tr>
<tr><td>Lab Slot</td><td>L 45+ L46</td></tr>
<tr><td>Registration No.</td><td>25BCE0553</td></tr>
<tr><td>Name:</td><td>Abdul Hadi</td></tr>
<tr><td>Mobile no.:</td><td>7594975428</td></tr>
<tr><td>Title::</td><td colspan="2">Hardware Interfacing of Led, Servo Motor, Ultrasonic Sensor and LCD with Arduino Uno</td></tr>
<tr><td>TRA Name:</td><td>Mahesh Kumar</td><td>Assigned date of Task: Sept 17, 2026</td></tr>
<tr><td>TRA Mobile number:</td><td>7594975428</td><td>Date of Completion: Oct 12, 2026</td></tr>
<tr><td>Faculty Name:</td><td>Dr. Debashish Dash</td><td>Assessment no: 8</td></tr>
<tr><td colspan="3">Email ID of student: abdul.hadi2025@vitstudent.ac.in</td></tr>
<tr><td colspan="3">VTOP uploading Date and time: 25/10/2026 11:59 pm</td></tr>
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Signature of TRA

Question 1:

Design a hardware circuit using a physical Arduino board for blinking a LED.

Aim of the Experiment:

To design a hardware circuit using a physical Arduino Uno board to blink an LED at an interval of 1 second.

Tools Required:

- Arduino Uno board
- USB cable (Type A to Type B)
- LED
- 220 Ω resistor
- Breadboard
- Jumper wires
- Computer with Arduino IDE

Brief Theory:

A Light Emitting Diode (LED) is a diode that emits light when it is forward biased, that is, when its anode (longer leg) is at a higher voltage than its cathode (shorter leg). An LED must always be connected in series with a current limiting resistor, otherwise excess current can damage the LED or the Arduino pin.
An Arduino digital pin set as OUTPUT supplies 5 V when it is HIGH and 0 V when it is LOW. The digitalWrite() function switches the pin between HIGH and LOW, and the delay() function holds the state for the given time in milliseconds. Repeating this switching in the loop() function makes the LED blink.

Algorithm:

1. Start.

2. Define the LED pin as digital pin 13.
3. Set the LED pin as OUTPUT in the setup function.
4. Turn the LED ON by writing HIGH to the pin.
5. Wait for 1000 ms.
6. Turn the LED OFF by writing LOW to the pin.
7. Wait for 1000 ms.
8. Repeat the process continuously.
9. Stop.

Embedded C Code with comments:

```
const int ledPin = 13;      // LED connected to digital pin 13

void setup() {
  pinMode(ledPin, OUTPUT);    // Set LED pin as output
}

void loop() {
  digitalWrite(ledPin, HIGH); // LED ON
  delay(1000);                // Wait for 1 second
  digitalWrite(ledPin, LOW);  // LED OFF
  delay(1000);                // Wait for 1 second
}
```

Manual calculation:

None

Procedure:

1. Connect the Arduino Uno board to the computer using the USB cable.
2. Open the Arduino IDE, select the board as Arduino Uno under Tools > Board and select the correct port under Tools > Port.
3. Place the LED on the breadboard.

4. Connect the 220 Ω resistor between digital pin 13 of the Arduino and the anode (longer leg) of the LED using jumper wires.
5. Connect the cathode (shorter leg) of the LED to the GND pin of the Arduino.
6. Type the program in the Arduino IDE and click Verify to compile it.
7. Click Upload to transfer the program to the Arduino board.
8. Observe the LED blinking ON and OFF at an interval of 1 second.

Output snips:



Result:

The hardware circuit was successfully built using the Arduino Uno board, and the LED blinked ON and OFF at an interval of 1 second.

Inference:

A digital output pin of the Arduino can be used to control an LED, and the blinking rate can be changed easily by changing the delay values in the program.

Question 2:

Design a basic hardware circuit for a servo motor which will rotate from 0 to 180 degrees.

Aim of the Experiment:

To design a basic hardware circuit using Arduino Uno and a servo motor that rotates from 0 to 180 degrees and back to 0 degrees.

Tools Required:

- Arduino Uno board
- USB cable (Type A to Type B)
- Servo motor (SG90)
- Jumper wires
- Computer with Arduino IDE

Brief Theory:

A servo motor consists of a DC motor, a gear system, a position feedback potentiometer and a control circuit. It rotates to a particular angle (0° to 180° for a standard hobby servo) and holds that position. A servo has three wires: power (red), ground (brown or black) and signal (orange or yellow).
The angle of a servo is controlled by the width of a pulse repeated every 20 ms (50 Hz). A narrow pulse of about 1 ms moves the shaft towards 0°, and a wider pulse of about 2 ms moves it towards 180°. The Arduino Servo library generates this signal on any digital pin when the angle is given using the write() function.

Algorithm:

1. Start.
2. Include the Servo library and create a servo object.

3. Attach the servo signal to digital pin 9 in the setup function.
4. Rotate the servo from 0° to 180° in steps of 1° with a delay of 15 ms after each step.
5. Rotate the servo from 180° back to 0° in steps of 1° with a delay of 15 ms after each step.
6. Repeat the process continuously.
7. Stop.

Embedded C Code with comments:

```
#include <Servo.h>          // Servo library

Servo myServo;               // Servo object
int angle;                   // Current angle

void setup() {
  myServo.attach(9);          // Servo signal on pin 9
}

void loop() {
  for (angle = 0; angle <= 180; angle++) {   // 0 to 180 degrees
    myServo.write(angle);     // Move to the angle
    delay(15);                // Wait for the servo to reach it
  }

  for (angle = 180; angle >= 0; angle--) {   // 180 back to 0 degrees
    myServo.write(angle);
    delay(15);
  }
}
```

Manual calculation:

None

Procedure:

1. Connect the Arduino Uno board to the computer using the USB cable.
2. Open the Arduino IDE, select the board as Arduino Uno and select the correct port.
3. Connect the red (power) wire of the servo to the 5V pin of the Arduino.
4. Connect the brown or black (ground) wire of the servo to the GND pin of the Arduino.
5. Connect the orange or yellow (signal) wire of the servo to digital pin 9 of the Arduino.
6. Type the program in the Arduino IDE and click Verify to compile it.
7. Click Upload to transfer the program to the Arduino board.
8. Observe the servo shaft rotating from 0° to 180° and back to 0° continuously.

Output snips:



Result:

The hardware circuit was successfully built, and the servo motor rotated from 0 to 180 degrees and back to 0 degrees continuously.

Inference:

The angle of a servo motor can be precisely controlled by the pulse width of the signal from an Arduino pin using the Servo library. A single micro servo can be powered directly from the 5V pin of the Arduino, but larger servos need a separate power supply with a common ground.

Question 3:

Design a basic hardware circuit for interfacing an ultrasonic sensor with Arduino Uno.

Aim of the Experiment:

To design a hardware circuit that interfaces an HC-SR04 ultrasonic sensor with Arduino Uno and to measure the distance of an object and display it on the serial monitor.

Tools Required:

- Arduino Uno board
- USB cable (Type A to Type B)
- Ultrasonic distance sensor (HC-SR04)
- Jumper wires
- Computer with Arduino IDE

Brief Theory:

The HC-SR04 ultrasonic sensor measures distance using sound waves. It has four pins: VCC, TRIG, ECHO and GND. When a pulse of at least 10 µs is applied to the TRIG pin, the sensor sends eight pulses of 40 kHz ultrasonic sound. When the sound is reflected back by an object, the ECHO pin goes HIGH for a time equal to the time taken by the sound to travel to the object and back.
The Arduino measures this time using the pulseIn() function. Since the sound travels to the object and back, the distance is calculated as distance = (time × speed of sound) / 2. The HC-SR04 works on 5 V and can measure from about 2 cm to 400 cm.

Algorithm:

1. Start.
2. Define the TRIG pin as digital pin 9 and the ECHO pin as digital pin 10.
3. Set the TRIG pin as OUTPUT, the ECHO pin as INPUT and initialize serial communication at 9600 baud.
4. Send a LOW signal on the TRIG pin for 2 µs and then a HIGH signal for 10 µs.
5. Set the TRIG pin LOW again.
6. Measure the time for which the ECHO pin stays HIGH using pulseIn().
7. Calculate the distance in centimeters.
8. Display the distance on the serial monitor.
9. Wait for 500 ms and repeat the process continuously.
10. Stop.

Embedded C Code with comments:

```
const int trigPin = 9;      // Trigger pin
const int echoPin = 10;     // Echo pin

long duration;              // Echo time in microseconds
float distance;             // Distance in cm

void setup() {
  pinMode(trigPin, OUTPUT);   // Trigger as output
  pinMode(echoPin, INPUT);    // Echo as input
  Serial.begin(9600);         // Start serial communication
}

void loop() {
  digitalWrite(trigPin, LOW);   // Clear trigger
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);  // Send 10 us pulse
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);    // Read echo time
  distance = duration * 0.0343 / 2;     // Calculate distance in cm
```

```
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(500);                    // Wait for 0.5 second
}
```

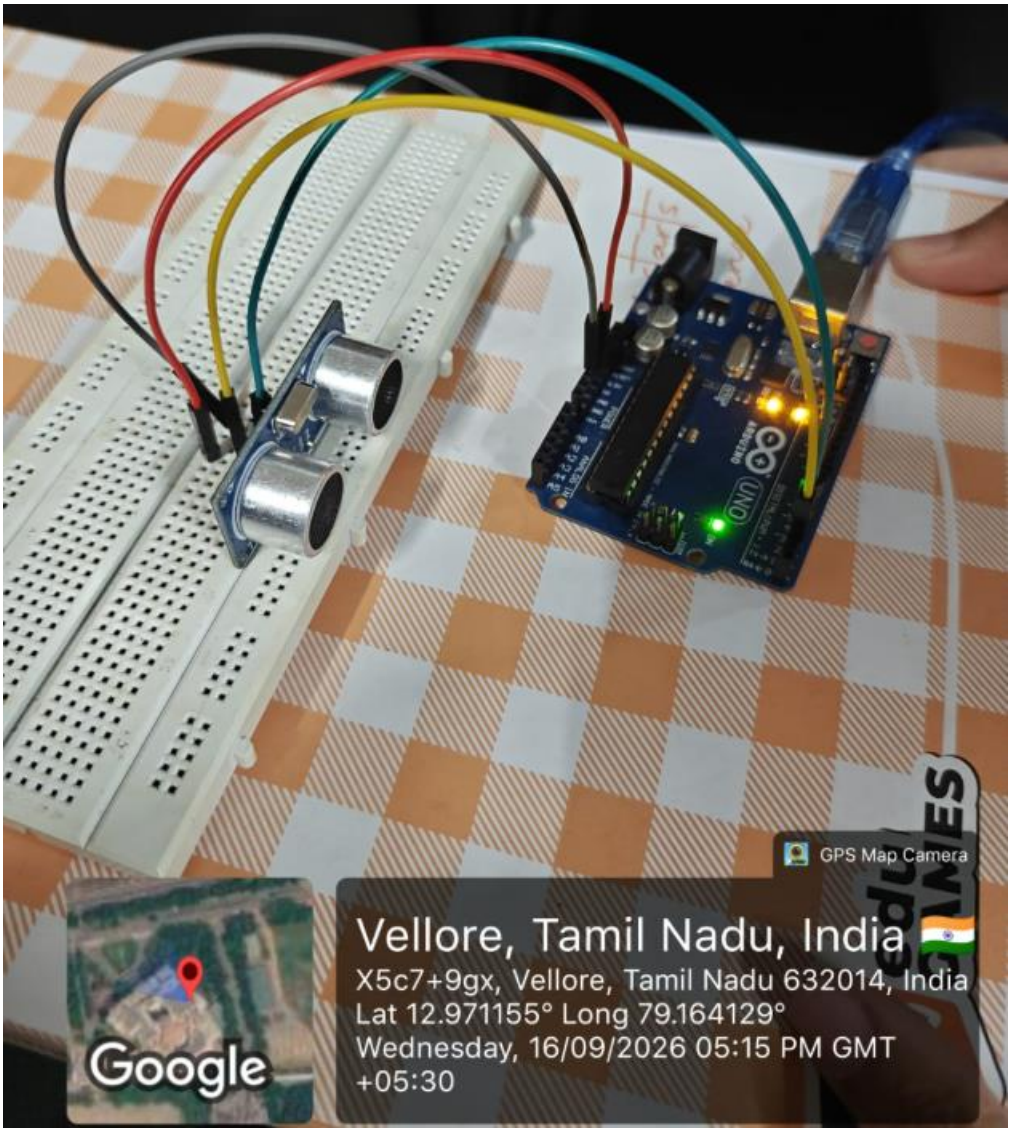
Manual calculation:

Speed of sound = 343 m/s = 0.0343 cm/µs
Distance = (Echo time × 0.0343) / 2

Procedure:

1. Connect the Arduino Uno board to the computer using the USB cable.
2. Open the Arduino IDE, select the board as Arduino Uno and select the correct port.
3. Connect the VCC pin of the HC-SR04 to the 5V pin of the Arduino.
4. Connect the GND pin of the HC-SR04 to the GND pin of the Arduino.
5. Connect the TRIG pin of the HC-SR04 to digital pin 9 of the Arduino.
6. Connect the ECHO pin of the HC-SR04 to digital pin 10 of the Arduino.
7. Type the program in the Arduino IDE and click Verify to compile it.
8. Click Upload to transfer the program to the Arduino board.
9. Open the Serial Monitor with the baud rate set to 9600.
10. Place an object in front of the sensor, vary its distance and observe the measured distance on the Serial Monitor.

Output snips:



Result:

The HC-SR04 ultrasonic sensor was successfully interfaced with the Arduino Uno board, and the distance of the object was displayed in centimeters on the serial monitor.

Inference:

The distance of an object can be measured accurately using an ultrasonic sensor by measuring the echo time with an Arduino, and the reading changes as the object is moved closer to or away from the sensor.

Question 4:

Design a hardware interfacing circuit for interfacing a normal 16×2 LCD as well as a potentiometer and print your name and registration number.

Aim of the Experiment:

To interface a normal 16×2 LCD and a potentiometer (for contrast control) with Arduino Uno and to display name and registration number on the LCD.

Tools Required:

- Arduino Uno board
- USB cable (Type A to Type B)
- 16×2 LCD (HD44780 compatible)
- 10 kΩ potentiometer
- 220 Ω resistor
- Breadboard
- Jumper wires
- Computer with Arduino IDE

Brief Theory:

A normal 16×2 LCD (HD44780 controller) can display 16 characters in each of its 2 lines and has 16 pins: VSS, VDD, V0, RS, RW, E, D0 to D7, A (backlight +) and K (backlight −). In 4-bit mode only the data pins D4 to D7 along with RS and E are connected to the Arduino, and the RW pin is connected to GND because the LCD is only written to.

The V0 pin controls the contrast of the display. A 10 kΩ potentiometer connected between 5 V and GND is used as a voltage divider, and its middle (wiper) pin is connected to V0, so turning the knob changes the contrast. The Arduino LiquidCrystal library is used to initialize the LCD, set the cursor position and print the text.

Algorithm:

1. Start.
2. Include the LiquidCrystal library and create the LCD object with RS = 12, E = 11, D4 = 5, D5 = 4, D6 = 3 and D7 = 2.
3. Initialize the LCD as a 16×2 display in the setup function.
4. Set the cursor to column 0 of line 1 and print the name.
5. Set the cursor to column 0 of line 2 and print the registration number.
6. Adjust the potentiometer until the text is clearly visible.
7. Stop.

Embedded C Code with comments:

```
#include <LiquidCrystal.h>    // LCD library

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);   // RS, E, D4, D5, D6, D7

void setup() {
  lcd.begin(16, 2);            // Initialize 16x2 LCD
  lcd.setCursor(0, 0);         // Column 0, line 1
  lcd.print("Abdul Hadi");     // Print name
  lcd.setCursor(0, 1);         // Column 0, line 2
  lcd.print("25BCE0553");      // Print registration number
}

void loop() {
  // Nothing to repeat, the text stays on the display
}
```

Manual calculation:

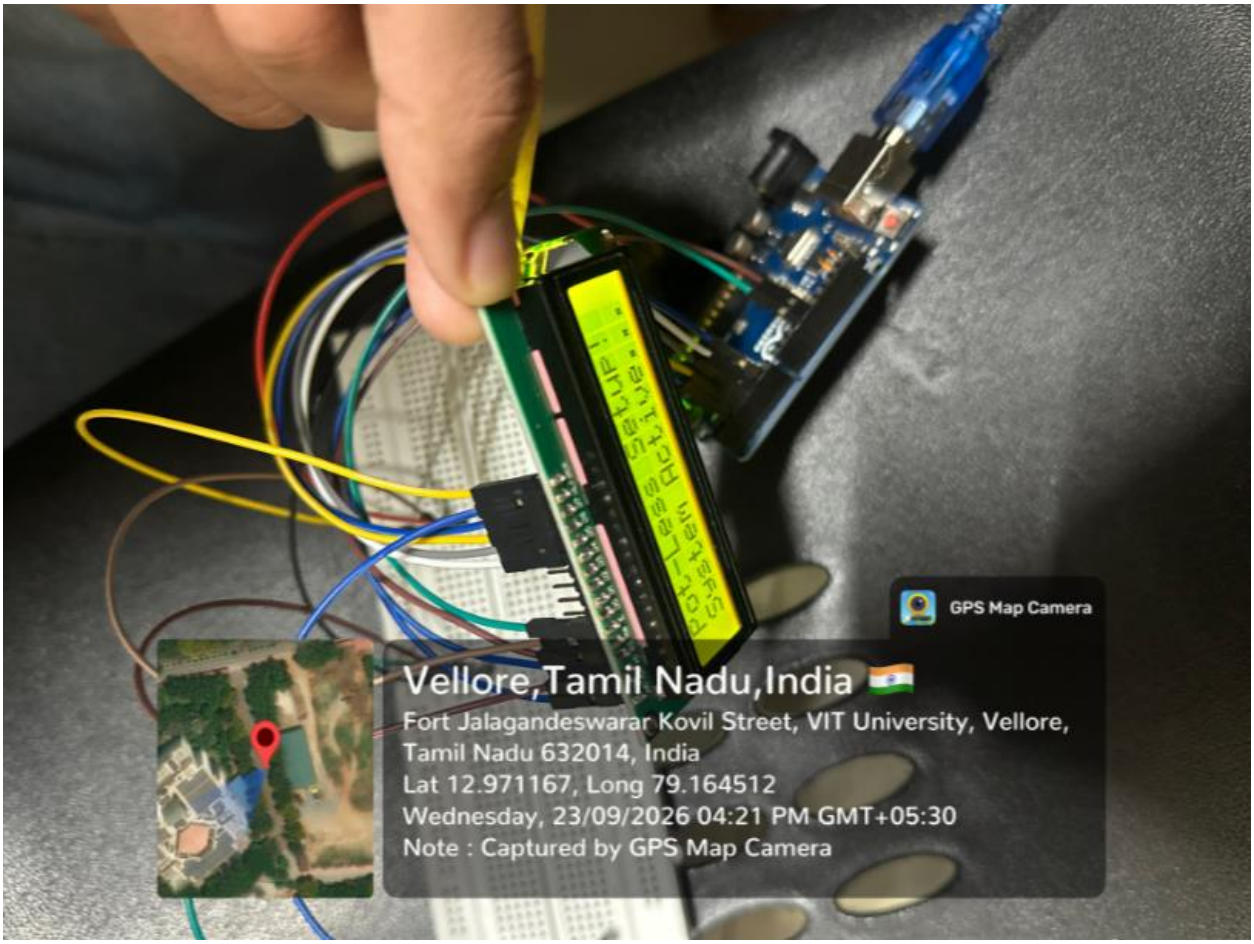
None

Procedure:

1. Connect the Arduino Uno board to the computer using the USB cable.

2. Open the Arduino IDE, select the board as Arduino Uno and select the correct port.
3. Connect LCD pin 1 (VSS) to GND and pin 2 (VDD) to 5V.
4. Connect the two outer pins of the potentiometer to 5V and GND, and its middle pin to LCD pin 3 (V0).
5. Connect LCD pin 4 (RS) to digital pin 12, pin 5 (RW) to GND and pin 6 (E) to digital pin 11.
6. Connect LCD pins 11, 12, 13 and 14 (D4, D5, D6 and D7) to digital pins 5, 4, 3 and 2 respectively.
7. Connect LCD pin 15 (A) to 5V through the 220 Ω resistor and pin 16 (K) to GND for the backlight.
8. Type the program in the Arduino IDE, click Verify and then click Upload.
9. Rotate the potentiometer knob until the contrast is correct and observe the name and registration number on the LCD.

Output snips:



Result:

The 16×2 LCD and the potentiometer were successfully interfaced with the Arduino Uno board, and the name and the registration number were displayed on the LCD.

Inference:

A 16×2 LCD can be interfaced with Arduino Uno in 4-bit mode using only six digital pins, and its contrast can be adjusted using a potentiometer connected to the V0 pin.

ESA Lab Assessment report

Sl.no.	Components	Marks Obtained
1	Pre Lab Work (Flow chart and Algorithm) documentation (2)	
2	Hardware or software Implementation (2)	
3	Manual calculation , Output data table / graph verification (2)	
4	Result with Inference (2)	
5	Post Lab Work (C Code with comments) documentation (2)	
	Total (10)	

1	Name: Abdul Hadi	
2	Reg. No.: 25BCE0553	
3	Lab slot: L45+L46	Lab. Slot Time: 3:50-5:30
4	Assessment no.: 8	
5	Title of Assessment: Hardware Interfacing of Led, Servo Motor, Ultrasonic Sensor and LCD with Arduino Uno	
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10	Email id: abdul.hadi2025@vitstudent.ac.in	
11	TRA Signature with Date	
12	Faculty Signature with Date	